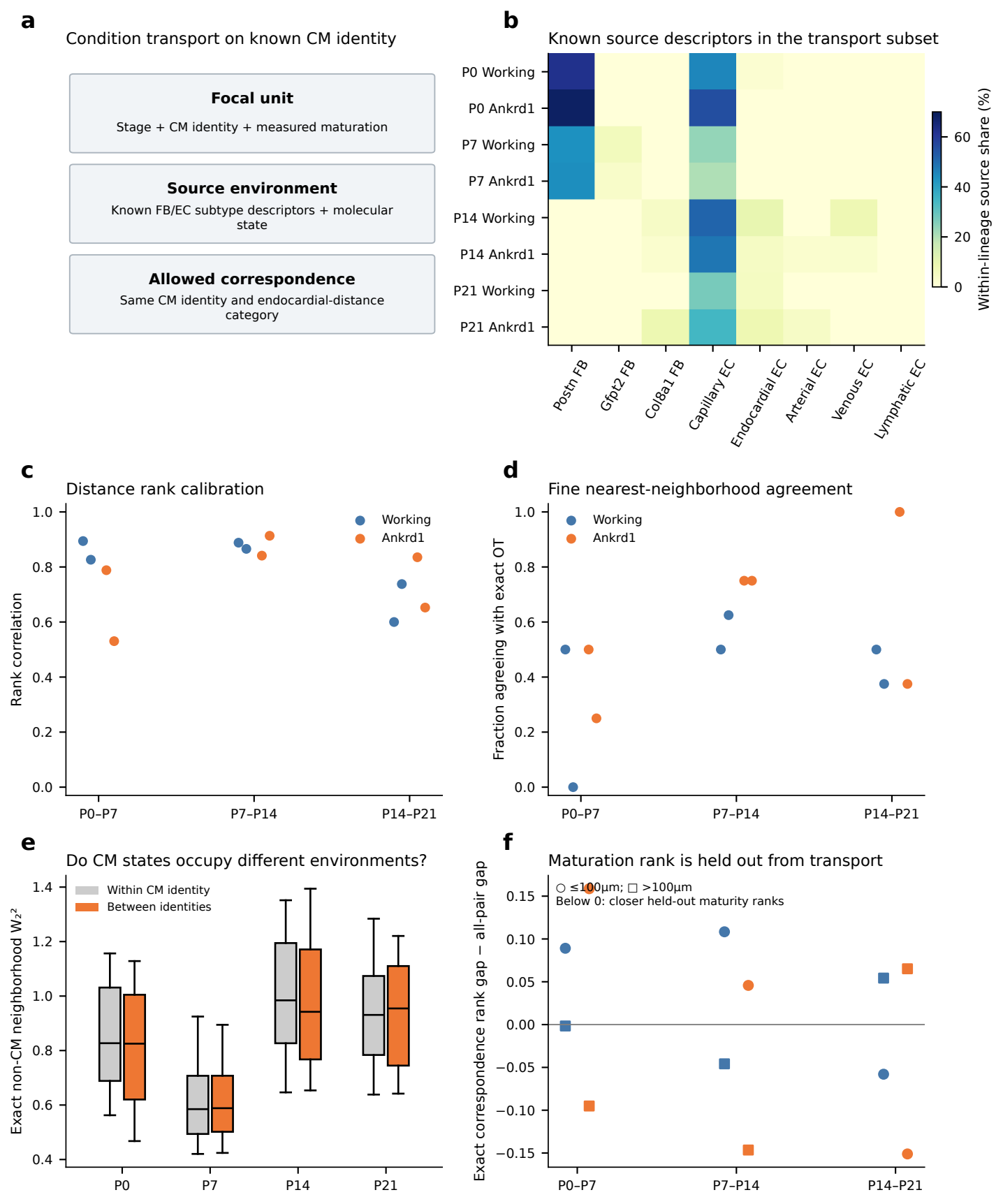
a
Condition transport on known CM identity
Focal unit
Stage + CM identity + measured maturation
Source environment
Known FB/EC subtype descriptors + molecular state
Allowed correspondence
Same CM identity and endocardial-distance category
b
Known source descriptors in the transport subset
P0 Working
P0 Ankrd1
P7 Working
P7 Ankrd1
P14 Working
P14 Ankrd1
P21 Working
P21 Ankrd1
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
Within-lineage source share (%)
0
20
40
60
c
Distance rank calibration
Working
Ankrd1
Rank correlation
0.0
0.2
0.4
0.6
0.8
1.0
P0–P7
P7–P14
P14–P21
d
Fine nearest-neighborhood agreement
Working
Ankrd1
Fraction agreeing with exact OT
0.0
0.2
0.4
0.6
0.8
1.0
P0–P7
P7–P14
P14–P21
e
Do CM states occupy different environments?
Within CM identity
Between identities
Exact non-CM neighborhood W₂²
0.4
0.6
0.8
1.0
1.2
1.4
P0
P7
P14
P21
f
Maturation rank is held out from transport
○ ≤100µm; □ >100µm
Below 0: closer held-out maturity ranks
Exact correspondence rank gap − all-pair gap
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
P0–P7
P7–P14
P14–P21